

Bitcoin Ordinals: Determinants and Impact on Total Transaction Fees

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Abstract

Bitcoin Ordinals are on-chain non-fungible assets native to the Bitcoin blockchain. As the data is stored on-chain, by opposition to Ethereum-like NFTs, creating an ordinal (i.e. “inscribing” it) can be expensive. We analyze the initial wave of Ordinals from December 14th 2022 until April 16th 2023. We show that ordinal inscriptions tend to have a lower fee rate than regular transactions. We also show that a high average block transaction fee rate is associated with lower ordinal inscriptions. These effects are larger for large inscriptions. This suggests that ordinal users are careful when setting the fee rate of their inscriptions. We also show that both the transaction fee rate for non-inscription transactions and total block fees have increased with the popularity of Bitcoin Ordinals, increasing miners’ revenue.

Keywords: Bitcoin ; Blockchain ; Ordinal ; Non fungible assets ; Transaction fees

JEL Codes: D23, G29, L14,

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1 Introduction

Ordinals are a way to add non fungible asset capabilities to the Bitcoin blockchain. Although Bitcoin is often opposed to Smart Contract platforms like Ethereum, Bitcoin developers have pushed the boundaries of regular Bitcoin transactions and found ways to allow the creation of a new type of asset that resembles Ethereum-based NFTs but lives on Bitcoin. Proposed in early 2022, the first ordinal has been inscribed on the Bitcoin blockchain on December 14th 2022.¹

The goal of this paper is threefold. First, we provide a description of Bitcoin ordinals, and details about their creation and lifecycle. Second, we analyze the recent trend in Bitcoin ordinals and explore the determinants of ordinal inscriptions. Lastly, we study the impact of Bitcoin ordinals on the Bitcoin blockchain, most importantly in terms of transaction fees and miners revenue.

Ordinals essentially refer to individual satoshis, the smallest possible sub-units of bitcoin, that can be uniquely identified and tracked when they are transferred. Also, a user can decide to send a particular satoshi, thereby allowing him to transfer an ordinal. Instead of having metadata on-chain and actual data off-chain like Ethereum-based NFTs, ordinals are fully operating on-chain and the data is directly written on the blockchain in a process called *inscription*. It is important to note that two technical upgrades on the Bitcoin network made ordinals possible, namely Segwit in 2017 and Taproot in 2021.² As ordinals essentially stored the data on-chain, inscriptions are usually costly because they can take a significant space within blocks, and therefore transaction fees will be of first importance throughout our analysis.³

We collect data on all ordinal inscriptions from the very first inscription in block 767,430 (mined on December 14th 2022) until block 785,651 (mined on April 16th 2023). Our dataset consists of 1,123,983 ordinal inscriptions. We also collect block level data. This data is by definition public and accessible to anyone running a Bitcoin full node or relying on a third-party node.

We show that, on average and in the cross-section, Bitcoin transactions with an inscription carry less fee rate than regular Bitcoin transactions from the same block.⁴ We also show that this transaction fee difference is even more pronounced for large ordinal inscriptions. This suggests that users are more careful with the fee

¹The process of “inscription” would corresponds to the “minting” process for Ethereum-based NFTs.

²Both were implemented as backward compatible upgrades. For more details see Pérez-Solà, Delgado-Segura, Herrera-Joancomartí, and Navarro-Arribas (2019), Brown, Chiu, and Koepl (2021).

³Even though the on-chain data receives the Segwit discount, so that it “counts” for only 25% of the actual size, this can still be significant.

⁴The transaction fee rate is usually measured in \$ per *virtual byte*, or *vbyte*. This is the relevant metric to consider because it corresponds to what miners are constrained when creating blocks.

rate they offer to miners which makes sense because the size of the inscription can yield large absolute fees.

We also find evidence that users time the market in terms of transaction fees, especially for large ordinal inscriptions. When the average transaction fees are high, blocks contain fewer ordinal inscriptions, especially large ones. This suggests that users may delay ordinal inscriptions when average transaction fees are high, especially when they are trying to inscribe large amounts of data. Our sample period corresponds to the early days of ordinals and even if there are periods of hype, no critical applications run on ordinals so inscription can usually be delayed.⁵

Our last result concerns the overall impact of ordinal inscriptions on the Bitcoin blockchain. We show that ordinal inscriptions have been associated with higher fees than non-inscription transactions. In addition, overall block transaction fees have also increased with ordinal inscriptions. This increase in total transaction fees is low relative to the current block subsidy award to miners from newly issued units of bitcoin, so the impact on consensus is marginal. However, this shows that it can have important implications in the future. Ordinals are a good example of applications that are built on the Bitcoin blockchain. Although they are trickier to develop than on Smart Contract platforms like Ethereum, Bitcoin can still run such applications. As more applications like this are built and used on Bitcoin, transaction fees should keep increasing, which is detrimental to consensus and therefore blockchain security.⁶

This paper is the first to provide an economic analysis of Bitcoin ordinals. Bitcoin ordinals have been introduced in Rodarmor (2022), and the first ordinal was inscribed on the Bitcoin blockchain on December 14th 2022. Kiraz, Larraia, and Vaughan (2023) presents a process to settle NFT trades on the Bitcoin blockchain, but this is inherently different from what ordinals achieve as explained in section 2.

This paper relates to the blockchain transaction fees literature. Bitcoin fees have been studied theoretically. For instance, Easley, O'Hara, and Basu (2019), Hinzen, John, and Saleh (2019) and Huberman, Leshno, and Moallemi (2021) have analyzed equilibrium transaction fees in Bitcoin. In different settings, they show that equilibrium fees are proportional to waiting time and therefore blockchain congestion.⁷ Although we do not directly study blockchain congestion nor equilibrium fees, we provide evidence of increased transaction fees for non-inscription transactions, which suggests that ordinal inscriptions have increased waiting time for all Bitcoin users.

⁵Note that an easy way to delay inscriptions is to set a lower fee. Indeed, an ordinal inscription, just like a regular Bitcoin transaction, will be broadcasted to the network. If the fee rate is lower than the current fee rate, the transaction will just sit in the mempool of pending transactions until transaction fees lower to a point at which it will make sense for a miner to include this inscription.

⁶Although it can sound sub-optimal, Bitcoin requires high total fees to operate properly in the long run.

⁷See Halaburda, Haeringer, Gans, and Gandal (2022) for a detailed literature review on Bitcoin and cryptocurrencies, with more insights on Bitcoin transaction fees.

Chiu and Koepl (2017) also use a model of endogenous transaction fees but in a different context. They study the optimality of miners' reward and show that rewarding miners with seigniorage is better and that transaction fees should be used as little as possible. In an early work, Möser and Böhme (2015) provide a description of transaction fees dynamics. We provide similar insights on the more recent period of early 2023.

It is important to note that most current papers only take the point of view of users willing to use Bitcoin as a payment system to study transaction fees. Ordinals are a completely different use of the Bitcoin blockchain and our paper shows that the impact on transaction fees can be quite large. This suggests that deeper theoretical analysis is required, and that competition in Bitcoin use cases may be an important determinant of transaction fees and security.

In this regard, our paper is also very much linked to the literature on Ethereum transaction fees. Indeed, in our paper we are able to directly identify a particular application on Bitcoin (i.e. ordinal inscriptions) and study their transaction fees compared to regular transactions. Donmez and Karaivanov (2022) study the economic determinants of transaction fees in Ethereum for all transactions and find similar evidence than the Bitcoin literature. They also show that transaction types are important and that a larger fraction of regular transfers is associated with higher fees. This is not consistent with our analysis. Indeed, we show that on average non-standard transactions (i.e. ordinal inscriptions) increase the transaction fees. This can be explained by the important difference between ordinals and Ethereum transactions. In an ordinal inscription, the data is included on-chain, increasing significantly the size of the transaction, and therefore the absolute transaction fee, while non-standard Ethereum transactions do not necessarily consume more gas than regular ones.⁸ To the best of our knowledge, there is no research on the difference in transaction fees for different type of Ethereum transactions. Note that Ethereum has undergone an upgrade in the transaction fee mechanisms that smooths out these differences anyway. With EIP-1559, the base fee is the same for all Ethereum transactions within a block and is a function of congestion. While users still have the possibility to include a priority fee, different Ethereum applications should not have significant differences in terms of transaction fees.⁹

The rest of the paper is organized as follows. Section 2 provides technical details on Bitcoin ordinals. Section 3 describes the data used in this paper. Section 4 analyzes the determinants of ordinal inscriptions. Section 5 provides insights on the overall impact of ordinals on the Bitcoin blockchain, and section 6 concludes.

⁸Obviously interactions with smart contracts can be complex, but pushing actual data on-chain, like with ordinal inscriptions, is orders of magnitude larger than regular transactions.

⁹Reducing fee volatility within a block was one of the objectives of EIP-1559. For an economic analysis of this change in fee mechanism, see Roughgarden (2020).

2 What are Ordinals ?

2.1 Ordinals

Ordinals are a type of Non-Fungible token (or NFT) native to the Bitcoin blockchain. Ordinals are implemented as an abstraction of Bitcoin transactions which means they do not require any update of the Bitcoin transactional system nor the consensus protocol.

The smallest subdivision of a bitcoin (*BTC*) is called a satoshi (or sat), denoted by *SAT* and represents 100 millionth of a bitcoin, $1 \text{ SAT} = 10^{-8} \text{ BTC}$. By definition, satoshis cannot be divided into smaller pieces and are natively non-fungible but the Bitcoin transactional system pools them in inputs, outputs and wallets. To create a system of non-fungible assets it is therefore enough to allow the tracking of individual satoshis.

Ordinal theory introduced by Casey Rodarmor in February 2022,¹⁰ is a standard for numbering, or “ordering”, and tracking individual sats on the Bitcoin blockchain. This is achieved with 3 main ideas/conventions :

1. Sats are associated with a serial number given the order they have in the coinbase transaction that created them. Ex: the i^{th} sats of the j^{th} block has a serial number of $\sum_{t=1}^{j-1} \pi_t + i$ with π_t the max number of sats that could have been created at block $t < j$.
2. For every transaction, sats in the inputs are passed on to sats in the outputs based on a first-in-first-out convention.
3. For every transaction, sats paid to miners as fees are paid after all the transaction’s output.

These 3 conventions are enough to enable the tracking of every single sats on the Bitcoin blockchain. Note these are just conventions, because users coordinate on this standard, it will work for all, but this also means that it would be possible to create another tracking system for example using a first-in-last-out convention for transactions that would enable a different tracking system. This is unlikely to be used as just one standard for non fungible assets is enough but this is to emphasize that ordinals change nothing on the Bitcoin blockchain and just uses the underlying transaction system in a separate way. To use ordinals, users need dedicated wallet software that allows them to send a specific sat to a given input. This is not a standard behavior of traditional wallets but some wallets are ordinal-ready.¹¹

¹⁰The Ordinal specification is described in a Bitcoin Improvement Proposal (BIP) draft in Rodarmor (2022).

¹¹For example see the Ordinals Wallet, Xverse Wallet, and the Sparrow Wallet.

2.2 Inscriptions

As ordinals are sats, which are themselves stored in Bitcoin unspent output (or UTXO), it is possible to write some arbitrary data to ordinals in a process called an *inscription*. Inscriptions are on-chain data associated with a given satoshi. Technically, data is inscribed in the *scriptpubkey* of a Bitcoin input in a two-step process with a commit-reveal structure. First, a user sends the desired sat as the first sat of an output to the hash of a script containing the data to be inscribed. Then the spend of this output is made in a follow-up transaction revealing the content of the script and therefore the inscription data. This process is pictured in diagram 1.

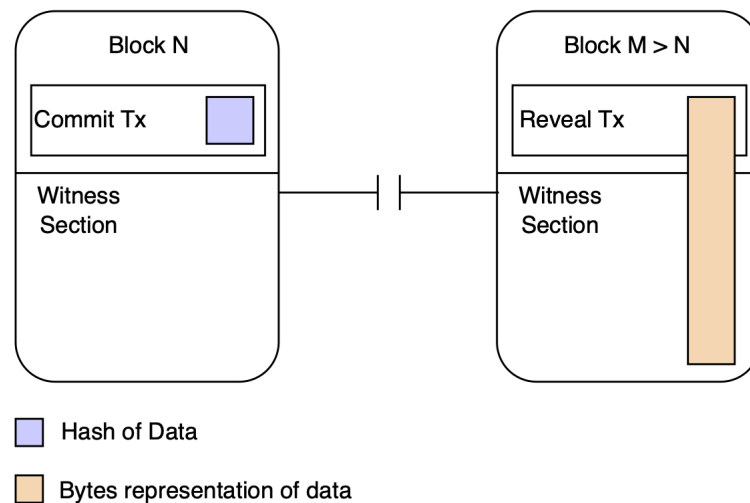


Figure 1: Commit-Reveal transaction schemes of ordinal inscriptions. First in a block N , the commit transaction sends some satoshis to a script containing the hash of the inscription data. The hash is effectively in the output and does not receive the segwit discount. Then, in a later block $M > N$, the reveal transaction spends the previous output, which can only be done by revealing the content of the inscription data. The data here receives the segwit discount because it is in the input script of a segwit transaction (more precisely a taproot transaction).

In the script of which the hash is used in the commit transaction, the inscription script is appended at the end of any other script. Inscriptions follow the content model of the web, so the inscription script contains the type of the inscription (or MIME type) and a byte string representation of the inscription data itself. Bitcoin Script contains special operation codes, grouped under the *op_push* code, making the script execution engine to push arbitrary data onto the execution stack. In order to facilitate standardization, inscriptions are serialized within an if-end statement called an envelope with the following structure :

```

OP_FALSE
OP_IF
  OP_PUSH "ord"
  OP_1
  OP_PUSH "image/jpeg"
  OP_0
  OP_PUSH 0x... (serialized byte data)
OP_ENDIF

```

Script 1: Bitcoin Script required to be pushed at the end of a Bitcoin Input to make an Ordinal inscription

This piece of script can be appended to any script because it does not change the stack.¹² Instructions inside the IF-END statements will never be executed but the full script contains valid instructions to be treated as standard by the Bitcoin protocol.¹³ Inside the IF statement, the first `OP_PUSH` determines that this is an ordinal, after the `OP_1` the type is specified and after `OP_0` the actual inscription data is specified.¹⁴ This creates a perfectly valid Bitcoin script that will be treated by ordinals-unaware nodes as a regular Bitcoin transaction.

Inscribing data directly on the blockchain can quickly become expensive. In Bitcoin, block space is the scarce resource for which users must compensate miners that select their transactions (see for instance Easley, O'Hara, and Basu (2019)). The main reason for the commit-reveal structure highlighted in figure 1 is to put the inscription script (and therefore the inscription data) in the signature part of the transaction hence benefiting from the witness discount associated with Segwit and Taproot's signatures. Segregated Witness, a soft fork activated in August 2017, changes the way block size is calculated. Before blocks had a maximum size of 1 MB each. After Segwit, the maximum block size is 4 million Weight Units (WU) which are a new units of account for transactions bytes : each byte in the witness data (including signatures and more generally input's *scriptpubkey*) accounts for 1 WU, while each byte outside of the witness data (everything else) accounts for 4 WU. Segwit is backward compatible because if all transactions use only non-witness data, the maximum block size is still 1 MB.¹⁵ Because the inscription data is in the witness part of a transaction, this means that they receive a 25% discount on the

¹²Indeed, `OP_FALSE` pushes a 0 to the stack, `OP_IF` pops an element off the stack and skip to after `OP_ENDIF` if this element is 0, which is always the case here.

¹³For an analysis of non-standard Bitcoin transactions, see Bistarelli, Mercanti, and Santini (2018).

¹⁴As mentioned, the type of ordinals is based on the format already used as *content-type* on the web, also called MIME types. A non-exhaustive list of MIME type can be found at https://developer.mozilla.org/en-US/docs/Web/HTTP/Basics_of_HTTP/MIME_types/Common_types.

¹⁵More detail on segwith can be found in Pérez-Solà, Delgado-Segura, Herrera-Joancomartí, and Navarro-Arribas (2019) and Antonopoulos (2019).

actual data size.

There is currently a theoretical limit to a Bitcoin inscription of 4 MB that would be achieved with only a single transaction in the block. In reality, some Bitcoin transactions, valid from the consensus point of view are considered non-standard and therefore may not be relayed by nodes and/or mined. This is when a transaction has a total weight of more than 400k WU. With this limit, it will not be considered standard to create an inscription of more than 400 kB.¹⁶

Let's take the example of a 20kB inscription on the blockchain. As can be seen on figure 2, a fee rate of 0.005 USD/vBytes is in the average of the recent blocks.¹⁷ To simplify, let's assume the inscription is 20k WU, that is we do not take into account the non-witness part of the transaction which is insignificant given the size of the inscription. With this assumption, the inscription will be 5k *vbytes*, and would cost $0.005 * 5000 = \$25$.

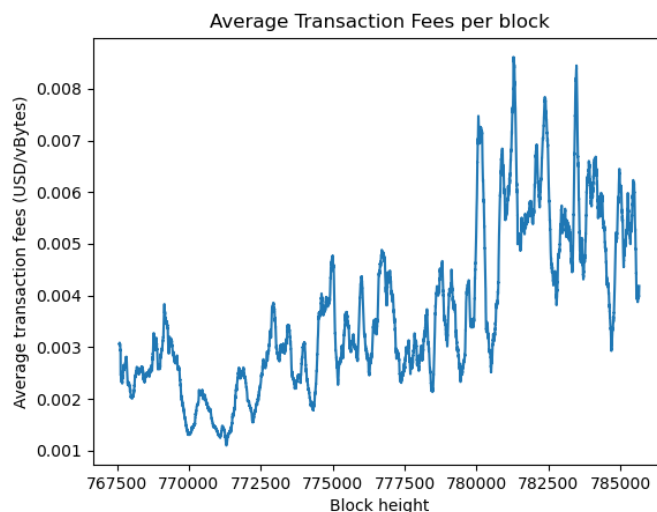


Figure 2: Average transaction fees in USD per *vbytes* for each block. The total fees in BTC of each block is divided by the quantity of *vbytes* (=25%WU) in the same block, and we then multiply this by an average of the bitcoin price between this block time and the previous block time. For clarity, this is the 144-block moving average.

By convention, an inscription is linked to the first satoshi of the first output of the reveal transaction (which contains the actual inscription). After this inscription, the satoshi, now called an ordinal, can be tracked using the ordinal concept presented above. The inscription on the blockchain, and therefore the inscription fees, are only paid once, and any subsequent transfer can be made just as a regular transaction.

Note that before ordinals and inscriptions, it was possible to write data on the Bitcoin blockchain using the `OP_RETURN` code. This code is heavily used for a num-

¹⁶See section 3.3 for information on very large inscriptions.

¹⁷In Bitcoin, fees are generally expressed as sats or USD per virtual bytes (or vBytes). A vByte is equal to 0.25 WU.

ber of use cases (see Strehle and Steinmetz (2020)), but it is much less convenient for at least 2 reasons. First, `OP_RETURN` is generally used in the output of a transaction, which means that the data written on-chain does not benefit from the witness discount, and is therefore 4 times more expensive than ordinal inscriptions. Second, when `OP_RETURN` is present in the output script, it makes the output non-spendable and this output is not even included in the UTXO set. With ordinals, it is possible to inscribe data on a satoshi and still be able to transfer this satoshi around, creating more use cases.

It is important to note that, although ordinals and inscriptions do not require a protocol change per se, they have been made possible with two recent protocol upgrades, namely Segwit and Taproot. Without being too technical, Segwit solves transactions malleability and allows for some part of the transaction to receive a discount in terms of the storage size it takes in a block, making it more accessible to inscribe data on the blockchain. Taproot changes the way Bitcoin Script is interpreted, releases a number of constraints on the Script, and allows for a fine selection of which satoshi to send to a given address. Both were implemented as backward compatible soft forks : Segwit was activated at block 481,824 on August 24th 2017 and Taproot was activated at block 709,632 on November 14th 2021.

2.3 Difference with other NFT object

Ordinals differ from the usual Ethereum-like NFT in two aspects.¹⁸ On Ethereum for instance, the vast majority of NFT represents digital images that are stored on decentralized servers like the Internet Protocol File System (IPFS), while only the id number, or *hash*, and some other metadata (like the IPFS link), is stored in a smart contract on-chain. When Alice buys a NFT, the smart contract changes the ownership of the entry associated to the bought NFT. For more details see Wang, Li, Wang, and Chen (2021).

Ordinals are radically different. First, ordinals inscribe all the data on-chain rather than off-chain. As previously explained, inscriptions are entirely written on-chain, in the signature part of a transaction. This makes this system not to rely on any protocol other than Bitcoin. It comes of course at the expense of higher fees, especially for large inscriptions. Note that even for a large inscription, the high transaction fees are paid only once at inscription time, and later transfers are regular-sized Bitcoin transactions.

The other difference is that ordinals need no management performed by smart contracts. On Ethereum-like NFTs, it is possible to add a number of features on

¹⁸EVM stands for Ethereum Virtual Machine, an EVM-like represents smart contract platforms more generally.

top of NFTs, including royalties and permissions, while ordinals are nothing more than a convention on top of regular Bitcoin transactions, entirely permissionless.

3 Data

3.1 Description

We collect public data on Bitcoin transactions using the dune.com service. Dune offers a way to query all Bitcoin blockchain data through a single entry point and a SQL-like query language.

We collect all transactions that have at least a Segwit input following the pattern described in Script 2.2 from the first one in block 767,430 (mined around 8:30pm UTC on December 14th 2022) until block 785,651 (mined around 11:50 UTC on April 16th 2023). For each of these transactions, we collect transaction level data as well as block level data corresponding to blocks containing these transactions.¹⁹ We also collect the bitcoin price at each block as served by dune.com. Their measure of the bitcoin price is computed as a volume-weighted price based on real-time data pulled from the Coinpaprika API.²⁰ From the data we collect, we construct several additional variables. Table 7 in appendix presents a description of all used variables.

3.2 Summary statistics

The dataset collected consists of all ordinal inscriptions made before block 785,651 (mined around 11:50 UTC on April 16th 2023). During that period there were a total of 1,123,983 inscriptions. Table 1 presents per-type inscription data. The *text* and *image* formats account for the vast majority, with more than 98% of all the inscriptions combined. There is a total of 2.39 GB of inscription data on the Bitcoin blockchain.²¹ Table 2 presents the summary statistics for some of the variables of interest.

While the first ordinal has been inscribed on December 14th 2022, there has been several waves of popularity. As can be seen in figure 3, there were almost no inscriptions for the first 7,500 blocks (i.e. 52 days). Then in February 2023, we started to witness hundreds of inscriptions per block.

¹⁹Note that, as mentioned in section 2.2, inscribing an ordinal takes actually two transactions : the commit and the reveal. Our dataset does not contain the commit transactions but only the “reveal” ones, with the actual data. Commit transactions are just like regular transactions and are usually small in size.

²⁰Coinpaprika API is available here : coinpaprika.com.

²¹At the end of our data sample period, on April 16th 2023, the total blockchain size was about 470 GB. Moreover note that as the inscribed data is in the signature part of blocks, it can be pruned once verified by nodes that do not care about ordinals.

Type	Number of inscriptions		Size of inscriptions	
text	556,909	49.55%	144 MB	6.03%
image	553,067	49.21%	2.17 GB	90.63%
application	11,715	1.04%	37 MB	1.56%
video	1,735	0.15%	32 MB	1.37%
audio	349	0.03%	9.5 MB	0.40%
other	115	0.01%	183 KB	<0.01%
model	93	<0.01%	458 KB	0.02%
Total	1,123,983	100%	2.39 GB	100%

Table 1: Ordinal inscriptions per inscription type from the first inscription until block 785,651.

Name	mean	min	25th	50th	75th	max
Transaction level						
Transaction size (in <i>vbytes</i>)	2,125	98	152	190	672	980,000
Transaction fees (in $10^{-3}\$/vbytes$)	2.13	0	1.13	1.72	3.54	5.26
Block level						
Block size (in <i>Mvbytes</i>)	1.68	<0.01	1.38	1.57	1.97	3.95
Block tx count (in 10^3)	1.99	<0.01	1.21	2.04	2.80	6.14
Av Block tx fees (in $10^{-3}\$/vbytes$)	3.64	0	1.80	2.94	4.90	97.49
Ordinal count	61.7	0	0	6	59	3,785
Ordinal text count	30.57	0	0	0	14	2,131
Ordinal non-text count	31.13	0	0	2	28	3,472
Av block transaction fees						
All transactions (in $10^{-3}\$/vbytes$)	3.63	0	1.80	2.94	4.90	97.49
Ordinals (in $10^{-3}\$/vbytes$)	3.18	0	1.18	2.33	4.61	60.56
Non ordinals (in $10^{-3}\$/vbytes$)	3.93	0	2.03	3.40	5.27	97.52
Ordinals text (in $10^{-3}\$/vbytes$)	3.66	0.18	1.69	3.08	4.84	73.63
Ordinals non-text (in $10^{-3}\$/vbytes$)	2.95	0	1.13	2.11	4.37	44.83

Table 2: Summary statistics of some of the main variables

Ordinals also greatly differ in size. As can be seen on figure 4, there are much more very small ordinal inscriptions, of a few *vbytes*, while some inscriptions are quite large. We have excluded from this distribution 3 outliers that will be discussed in section 3.3, but there are still some ordinal inscriptions close to 100KvB, which is 4 times more in terms of actual *bytes*. Inscribing such large ordinals can be expensive, and we will see in the next section whether agents time transaction fees to make such large inscriptions.

3.3 Outliers

Within this dataset there are a number of outliers that are worth discussing. Figure 5 presents the size of ordinal inscriptions in order they appear on the blockchain. We clearly see that 3 transactions have inscribed data over 3MB. Given that the total

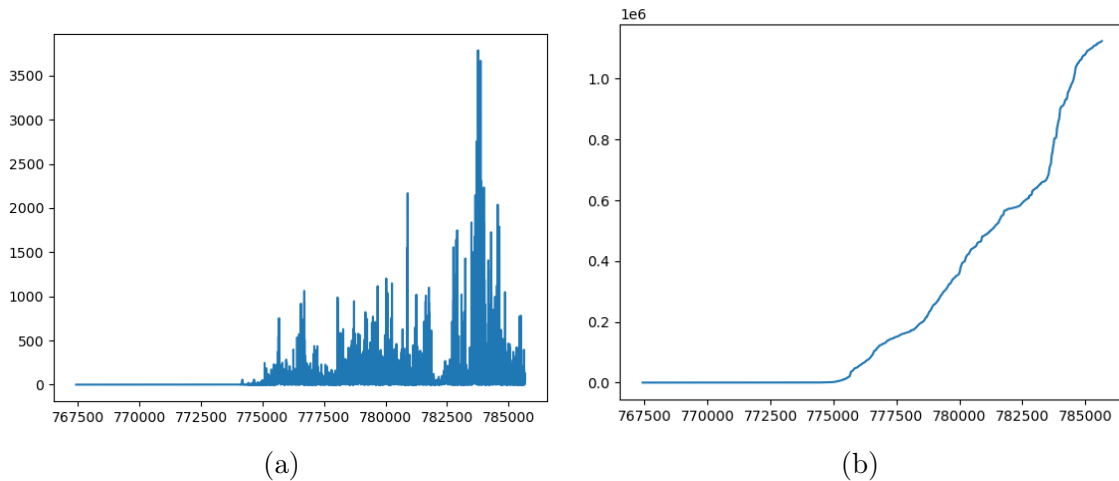


Figure 3: Ordinal inscriptions over time. Panel (a) presents the number of inscription per block over the whole sample period. Panel (b) presents the cumulative number of ordinal inscriptions over the whole sample period.

block size, including the signatures, is 4MB, this represents almost a full block.²²

Table 3 presents some metrics on these 3 outliers. We can be sure that one of them was actually included by a miner itself as it has no transaction fees.²³ We also see that \$18,178 and \$12,323 were spent to inscribe an image and a video respectively.²⁴ Note that these images and the video are high resolution ones, hence their size, but they are not really high quality images and could easily have been compressed down. This would have saved space on the blockchain and would have been cheaper to inscribe. This can be related to the meme economy in which participants tend to not act rationally but rather guided by some non-financial reward they obtain from the community for being extreme in their behavior.

For the rest of the analysis we have removed these 3 inscriptions from our dataset, and leave this for further studies.

4 The Determinants of Inscriptions

We are interested in understanding the determinants of ordinal inscriptions. Larger inscriptions are more expensive to make, at a given point in time, but transaction

²²The block size is 1MB, but is also 4WU which corresponds to 4MB blocks if we account for the segwit part of the block, and all transactions in that block use segwit.

²³The other ones could also come from miners as fees are neutral for them. To verify this, we would need to check the mempool data, to which we do not have access, to see if and when these transactions were received by the network. Also, as mentioned in section 2.2 forwarding transactions larger than 400kB is considered non-standard for most nodes, so it is unlikely that a non-miner would deal with these transactions.

²⁴Even if miners themselves included these transactions, this cost is an opportunity cost what they could have earned by including regular user transactions.

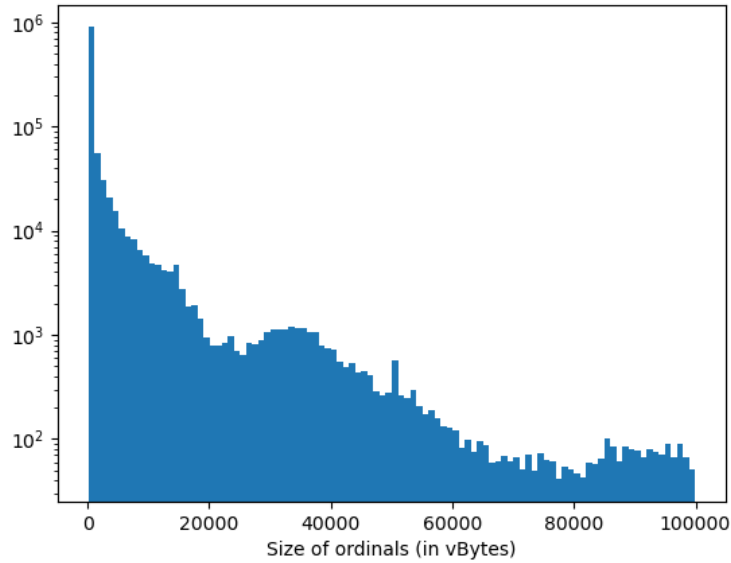


Figure 4: Distribution of the ordinal inscriptions' sizes, in log scale. We excluded 3 outliers with respective size 849,930 *vbytes*, 975,440 *vbytes* and 984,667 *vbytes*, which accounts for almost a full block (limited to 1MvB).

Tx hash	Tx size (in MB)	Tx fees (in \$)	Block height	% of block	Nb of Tx	Type
0x79b91...	3.90	18,178	777,945	99.2%	98	Image
0xb5a7e...	3.40	12,323	776,884	94.4%	446	Video
0x0301e...	3.94	0	774,628	99.6%	63	Image

Table 3: Descriptive data on outliers in terms of inscription size. The 5th column represents the percentage of the block that is taken by this ordinal inscription. The 6th column represents the number of transactions in this block.

fees per *vbyte* are time-varying and depend on the blockchain congestion.²⁵

4.1 Cross-section analysis

We first analyze the difference in fees among inscriptions of the same block. As inscriptions are less critical than regular transactions, agents may be willing to set relatively lower fees than regular Bitcoin transactions. Indeed, getting an ordinal inscribed does not usually carry a time emergency, while users may generally face a delay cost when submitting a regular transaction. At the same time, since ordinals are usually large transactions the total fee can be quite large, so trying to pay as little as possible may be more economically important for ordinal inscriptions than

²⁵A virtual byte, or *vbyte*, is the correct unit to measure size on the Bitcoin blockchain. See section 2.2 for more details.

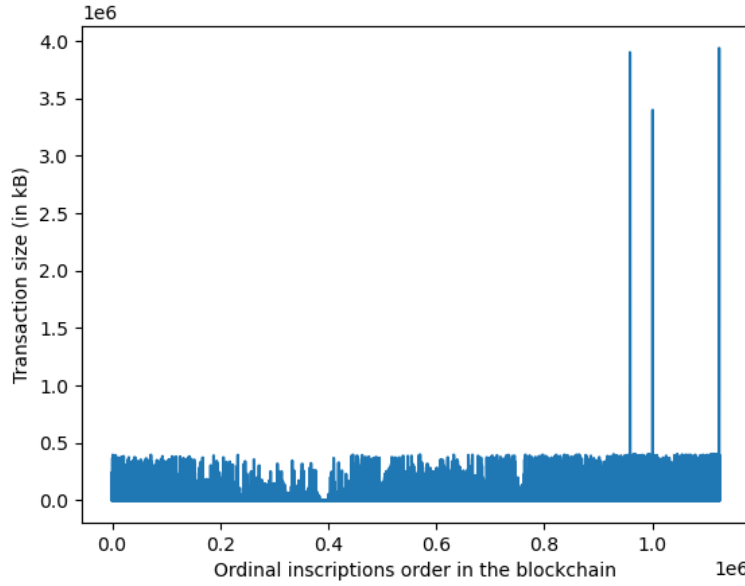


Figure 5: Transaction size of all ordinal inscriptions. The x-axis is the order of appearance in the blockchain.

regular transactions. We summarize this idea in the following hypothesis.²⁶

H1: *Bitcoin transactions with an inscription carry less fees than regular Bitcoin transactions from the same block.*

To test for this hypothesis, we compute the average transaction fee associated to ordinal inscriptions, and the one associated to other transactions for each block, and compare them at the block level. Then we test the paired-mean difference and present statistics about this difference.

Let's denote by f_b^o the average transaction fee for all the ordinal inscriptions in block b , and by f_b^{no} the average transaction fee for all non inscription transactions in block b , both measured in USD/vbyte. We then construct the series Δf_b^{o-no} defined by

$$\Delta f_b^{o-no} = f_b^o - f_b^{no}.$$

To strengthen our point, we consider only blocks that have at least 5 ordinal inscriptions. Table 4 presents statistics about Δf_b^{o-no} as well as the Wilcoxon signed-rank test which is a non parametric hypothesis test to compare two samples.²⁷ We also present statistics for the relative difference, F_b^{o-no} , defined as

$$F_b^{o-no} = \frac{\Delta f_b^{o-no}}{f_b^{no}}.$$

²⁶Throughout the paper, transaction fees are measures in terms of USD per vbyte.

²⁷We use the Wilcoxon test instead of the paired Student's t-test because we cannot reject the non-normal distribution hypothesis of the differences. Figure 6 also demonstrates the non-normality of the difference.

We see that the mean difference is negative and highly statistically significant according to the Wilcoxon test (low p-value). In economic terms, the relative mean difference is the relevant measure to consider. Ordinal inscriptions pay on average 39% less than non ordinal inscriptions of the same block. Figure 6 represents the distribution of Δf_b^{o-no} . We therefore validate hypothesis **H1**.

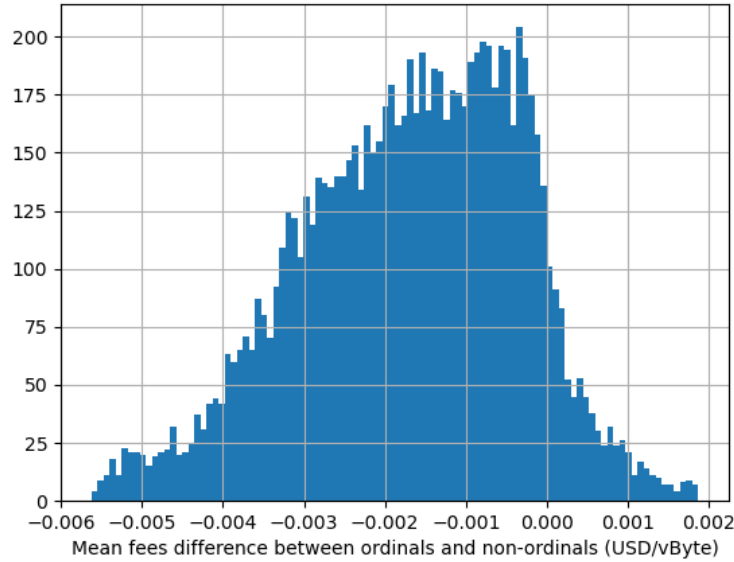


Figure 6: Distribution of the difference between average transaction fees of inscriptions minus non inscription transactions, F_b^{o-no} . We consider only blocks with at least 5 inscriptions.

To further extend this analysis, we now investigate whether the previous difference in fees comes from all types of ordinals or from only sub-groups. As observed in section 3.2, the vast majority of ordinal inscriptions is composed of the types text and images. Of all the different types, transactions inscribing text are by definition smaller than other inscriptions. Because the total fees paid depend on the size of the inscription, we expect large ordinal inscriptions to carry less relative fees. Because the type of transactions can be used to form groups of inscriptions by size, we investigate the difference in fees between text and non-text inscriptions. We summarize this idea in the following hypothesis.

H2: *Bitcoin transactions with a non-text inscription carry less fees than transactions with a text inscription from the same block.*

To test for this hypothesis, we use the same methodology as for **H1**. We compute the average difference in transaction fees between transactions with a non-text inscription and those with a text inscription, and compare them at the block level.

Let's denote by f_b^t the average transaction fee for the text inscriptions in block b , and by f_b^{nt} the average transaction fee for non-text inscriptions in block b , both

measured in USD/vbyte. We denote the relative difference by Δf_b^{nt-t} , such that

$$\Delta f_b^{nt-t} = f_b^{nt} - f_b^t$$

We only consider blocks with at least 5 inscriptions, and at least one inscription of each group (text and non-text). Table 4 presents statistics about Δf_b^{nt-t} as well as the Wilcoxon signed-rank test. We also present statistics for the relative difference, F_b^{nt-t} , defined as

$$F_b^{nt-t} = \frac{\Delta f_b^{nt-t}}{f_b^t}.$$

We see that the difference is negative and highly statistically significant. In economic terms, non-text inscriptions carry on average 16% less transaction fees. While this difference is economically meaningful, it is also lower than the difference between inscription and non inscription transactions. Figure 7 represents the distribution of Δf_b^{nt-t} . We therefore validate hypothesis **H2**.

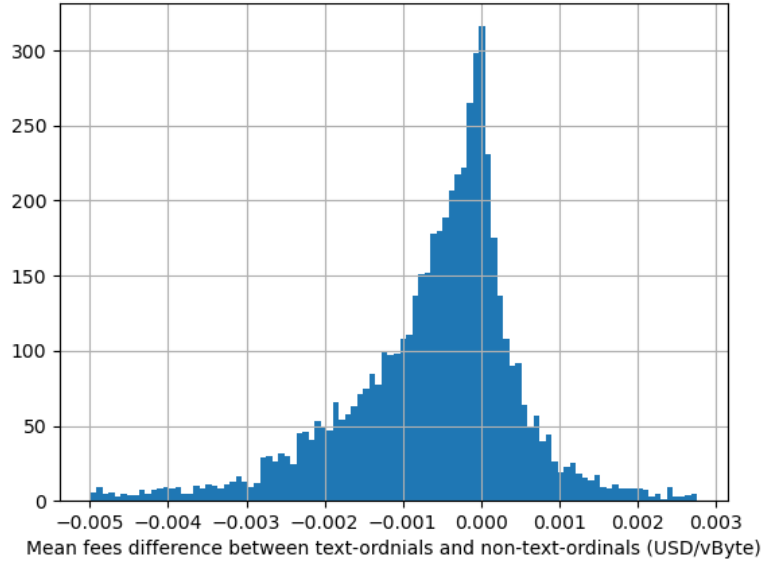


Figure 7: Distribution of the difference between average transaction fees of non-text inscription minus text inscription transactions, F_b^{nt-t} . We consider only blocks with at least 5 inscriptions, and at least one inscription in both groups.

4.2 Regression analysis

We now investigate the determinants of ordinal inscriptions per block. The number and size of inscriptions per block has increased substantially since ordinals inception because of exogenous factors such as the hype around ordinals, which is not directly measurable. Instead, to investigate the determinants of ordinal inscriptions, we relate to measurable quantities like the average block transaction fee.

Series	mean	min	25th	50th	75th	max	Wilcoxon-stat (p-value)
Δf_b^{o-no} (in 10^{-3} USD/vbytes)	-1.7	-5.6	-2.6	-1.6	-0.63	1.9	1221320 (0.0)
F_b^{o-no}	-0.39	-0.96	-0.64	-0.39	-0.17	3.19	N/A
Δf_b^{nt-t} (in 10^{-3} USD/vbytes)	-0.6	-5	-1.1	-0.4	0.02	2.8	2125417 (0.0)
F_b^{nt-t}	-0.16	-0.94	-0.41	-0.16	0.01	3.36	N/A

Table 4: Statistics for the difference in average transaction fees for different groups of transactions. Δf_b^{o-no} represents the absolute difference between ordinal inscriptions and regular transactions. F_b^{o-no} represents the same difference in relative terms. It contains only blocks with more than 5 inscriptions. Δf_b^{nt-t} represents the absolute difference between non-text inscriptions and text inscriptions. F_b^{nt-t} represents the same difference in relative terms. It contains only blocks with more than 5 inscriptions and at least one text and one non-text inscription.

When taking the decision to inscribe an ordinal, users may time transaction fees and delay inscription when transaction fees are high, which will result in less total inscriptions volume when fees are high. Under the assumption that users time overall block transaction fees, this effect should be even more pronounced for large inscriptions (i.e. non-text inscriptions).

In Bitcoin, the average block time is about 10 minutes. So between the block at which a user makes the decision to broadcast a transaction and the block at which the inscription is included in the blockchain, there can be several blocks. To investigate this, we propose a lagged linear regression model with the total size of ordinals per block as dependent variable and with independent variables being the average transaction fee in a block, and some lags of this value. We denote by V_b the size of all ordinal inscriptions in block b , by V_b^T the size of all text ordinal inscriptions in block b , and by V_b^{NT} the size of all non-text ordinal inscriptions in block b , as measured in *vbytes*. We also denote by f_b the average transaction fee in block b , measured in USD/vbyte. As observed in section 3.2, the number, hence the size, of ordinal inscriptions per block contains a significant positive trend. To avoid inference problems, we restrict the test sample to blocks containing at least 1 ordinal inscription (effectively removing all the blocks with zero inscriptions in the beginning of the sample), and consider the first difference of both the total size of ordinals and the transaction fees. We therefore propose the 3 following linear models:

$$\Delta Y_b = \alpha + \sum_{i=0}^6 \beta_i \Delta f_{b-i} + \epsilon_b \quad (1)$$

where $Y_b \in \{V_b, V_b^T, V_b^{NT}\}$ and $\Delta X_b = X_b - X_{b-1}$. Given the intuition exposed above, we formulate the following two hypotheses.

H3: *The relationship between transaction fees and total size of ordinal inscriptions should be negative, i.e. $\forall i \in 0, \dots, 6; \beta_i < 0$.*

H4: *The negative relationship between transaction fees and total inscription size should be more pronounced to non-text inscriptions compared to text inscriptions.*

To test these hypothesis, we run the OLS models (1) and present the results in table 5. To maintain regression coefficient in an easy-to-read range the dependent variables have been scaled down by 10^4 . Regression models (1) and (2) consider the total size of ordinal inscriptions for a given block. The change in average block transaction fees is negatively associated with the total quantity of ordinal inscriptions. In the model with 6 lags, all variables are negative and statistically significant, although their economic significance decrease with lag duration. To analyze economic significance, let's consider a coefficient of 5,000, which is close to the main coefficient in model (1). As the variables have been scaled, this means that an increase in average transaction fees by $\$0.001/vbyte$ translates to a decrease in total ordinal inscriptions by $0.001 \times 10^4 \times 5,000 = 50,000 \text{ vbytes}$. This would be even more if we consider a drift in average transaction fees with other lags being significant. We therefore validate hypothesis **H3**.

To further analyze this relationship, we account separately for the quantity of ordinal inscriptions of text and non-text inscriptions per block, measured in *vbytes*. Results are exposed in models (3) to (6) of table 5. We observe that the economic and statistical significance of the coefficient are similar for the non-text inscriptions, as they were for the total inscriptions, while they are much lower for text inscriptions. This suggests that users willing to inscribe non-text objects (i.e. a proxy for large objects) pay much more attention to transaction fees than for text (usually much lighter) inscriptions. Although the change in transaction fees of the last block significantly affects the total quantity of text inscriptions, the economic effect is about 27 times less in magnitude. To allow a comparison with models (1) and (2), let's compute the economic magnitude of the effect of a change in transaction fees on text inscriptions. If we consider a coefficient of 180, we have that an increase in average transaction fees by $\$0.001/vbyte$ translates to a decrease in text ordinal inscriptions by $0.001 \times 10^4 \times 180 = 1,800 \text{ vbytes}$. We therefore validate hypothesis **H4**.

5 Structural change in the Bitcoin Blockchain

The previous analysis has shown that within a block, ordinal inscriptions tend to have lower fees than non-inscription transactions, and that higher overall transaction

	All (1)	All (2)	Text (3)	Text (4)	Other (5)	Other (6)
$\alpha(\text{ const})$	17.86 (2262.01)	21.41 (2248.46)	1.13 (394.92)	1.12 (394.78)	16.72 (2242.70)	20.29 (2229.01)
Δf_b	-4977.59*** (82.55)	-5378.52*** (90.68)	-184.22*** (14.41)	-176.80*** (15.92)	-4793.37*** (81.84)	-5201.72*** (89.90)
Δf_{b-1}		-918.55*** (102.56)		34.23* (18.01)		-952.78*** (101.68)
Δf_{b-2}		-1002.59*** (104.76)		-12.66 (18.39)		-989.93*** (103.85)
Δf_{b-3}		-354.79*** (109.26)		26.55 (19.18)		-381.34*** (108.31)
Δf_{b-4}		-350.45*** (108.83)		-3.72 (19.11)		-346.73*** (107.88)
Δf_{b-5}		-284.30*** (103.38)		9.73 (18.15)		-294.03*** (102.48)
Δf_{b-6}		-170.31* (90.02)		-3.47 (15.81)		-166.84* (89.24)
R-squared	0.26	0.27	0.02	0.02	0.25	0.26
R-squared Adj.	0.26	0.27	0.02	0.02	0.25	0.26
N	10171	10171	10171	10171	10171	10171
F-stat	3635.97	544.13	163.39	25.18	3430.13	514.81
p-value	0.00	0.00	0.00	0.00	0.00	0.00

Table 5: OLS regression in first difference at the block level. For the first two models, the dependent variable is the total size of all inscriptions in a block, measured in *vbytes*. For the next two models, the dependent variable is the total size of text inscriptions in a block, measured in *vbytes*. For the last two models, the dependent variable is the total size of non-text inscriptions in a block, measured in *vbytes*. The independent variables are the average transaction fees in a block, over all transactions, measured in USD/*vbytes* as well as 6 lags of this variable. All independent variables have been scaled up by 10^4 to avoid too large coefficients. The sample is restricted to blocks containing at least one inscription. Standard errors in parentheses. The significance levels are : ***=1%, **=5%, *=10%.

fees are associated with lower inscription size. We now study the impact of the ordinal and inscriptions wave on the Bitcoin ecosystem by looking at fees on non-inscription transactions and total transaction fees.

5.1 Impact on non-ordinal transaction fees

As we have seen, ordinal inscriptions take space on the blockchain thereby increasing congestion and potentially impacting transaction fees of non-ordinal transactions.

To make this assessment, we compute the average USD transaction fees in each block but only for non-inscription transactions. Figure 8 shows that different regimes are indeed observable. As more inscriptions started to be included in the blockchain, transaction fees for non-inscription transactions increased, suggesting a negative externality to non-ordinal users.

To confirm the previous intuition, table 6 shows a numerical analysis. We see that we indeed observed significant breaks in terms of the average transaction fees for non-inscriptions transactions. The three estimated periods correspond to different

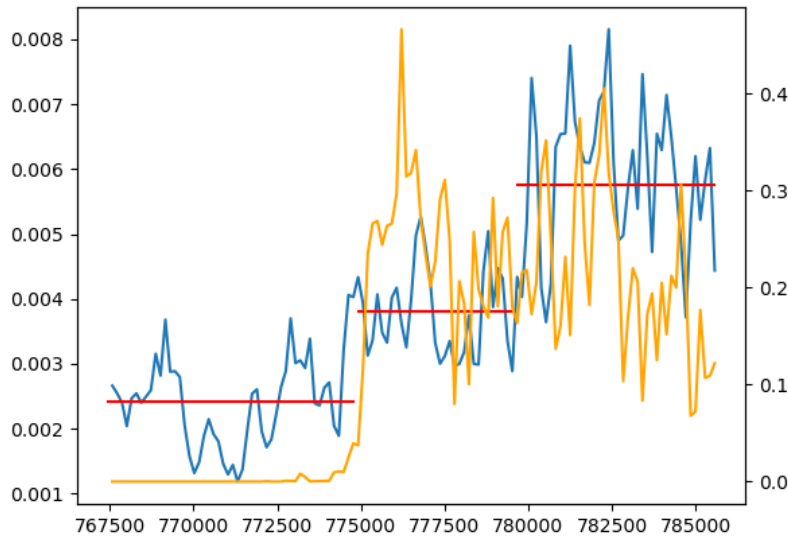


Figure 8: Average fees paid by non-inscription transactions in USD (in blue) and average proportion of blocks filled with ordinal inscriptions (in orange). We define 3 periods. The period (1) starts at block #767,430 (mined on 2022/12/14), the period (2) starts at block #774,775 (mined on 2023/02/02) and the period (3) starts at block #779,527 (mined on 2023/03/06). The red horizontal lines corresponds to the average fee paid by non-inscription transactions during the 3 periods. To smooth out the representation, we took the 144-block rolling window, that is the daily average.

periods in the history of ordinal inscriptions. The first period spans from early December 2022 until early February 2023 and corresponds to the early days with barely any ordinal inscriptions. The second period spans from early February 2023 until early March 2023 and corresponds to the first boom of ordinal inscriptions. The last period spans from early March 2023 until April 2023 and corresponds to a phase with less growth but a sustained proportion of ordinal inscription into the blockchain. For each of these subsequent periods the average transaction fee on non-ordinal inscriptions has kept increasing. This shows evidence of an overall positive impact on transaction fees of non-inscription transactions. As expected, ordinal inscriptions have increased congestion on the blockchain and other transactions had to pay more fees on average to get included in the blockchain. Between the first period and the last one, fees for non-inscription transactions have been multiplied by 2.39.

5.2 Impact on total transaction fees

Overall fees and blockchain security are intrinsically linked. Security can broadly be proxied by hashrate which is influenced by both block reward and global transaction fees.²⁸ Let's then explore whether total Bitcoin transaction fees have been affected

²⁸See Bertucci, Bertucci, Lasry, and Lions (2022), John, O'Hara, and Saleh (2022) or Pagnotta and Buraschi (2018) for more insights regarding the link between Bitcoin transaction fees and security.

	Period			(2)-(1)	(3)-(2)
	(1)	(2)	(3)		
Start Block	767,430	774,775	779,527		
End Block	774,775	779,527	785,576		
Average tx fees non-inscription (in 10^{-3} USD/ <i>vbytes</i>)	2.41	3.80	5.75	1.39*** (0.00)	1.95*** (0.00)
Average total block fees (in 10^3 USD)	1.94	3.34	5.23	1.40*** (0.00)	1.88*** (0.00)

Table 6: Impact of ordinal inscriptions on transaction fees for both 1) only non-inscription transactions, and 2) all transactions. We define 3 periods. The period (1) starts at block #767,430 (mined on 2022/12/14), the period (2) starts at block #774,775 (mined on 2023/02/02) and the period (3) starts at block #779,527 (mined on 2023/03/06). The two last columns correspond to the difference between the second and first periods and between the third and second periods respectively. Significance is tested with sample t-test, t-stat shown parentheses. The significance levels are : ***=1%, **=5%, *=10%.

by the ordinals movement.

So far we have shown that 1) ordinal inscriptions tend to pay lower fees than other Bitcoin transactions and that 2) the fees of non-inscription transactions has increased due to the ordinal movement. We now look at the effect on the total transaction fees to assess the impact on security and consensus. In table 6 we see that the effect is similar to the effect on non-inscription transactions. Miners have on average earned an additional \$3,280 per block, corresponding to a 2.7x increase. This is still marginal compared to the current block subsidy which is 6.25 BTC across our sample period, corresponding to around \$140,000 with a \$23,000 price.

The impact on consensus and on security is therefore marginal. Nonetheless ordinals are interesting in that they put a positive pressure on the fees, both stimulating development and use of other scaling techniques for regular transaction, as well as giving miners more incentives to mine from the fees, thereby increasing the blockchain security.

6 Conclusion

We have provided an analysis of the initial wave of Bitcoin ordinals. Although relatively new, Ordinals are an important step in the history of Bitcoin. While Bitcoin was long thought of as a rather low and inefficient payment system, ordinals show that it can also become a digital asset platform similar to more complex Smart Contract platforms like Ethereum.

We have shown that ordinal inscriptions usually carry less fees than non-inscriptions in the same block. Our results further extend to show that a high average block transaction fee rate is associated with fewer ordinal inscriptions. These effects are

even more pronounced for large ordinal inscriptions. This suggests that users are particularly careful when inscribing ordinals. This was expected because total fees can be very high with large inscriptions, i.e. large transactions.

We also show evidence of a positive correlation of Bitcoin ordinal inscriptions on transaction fees of other transactions as well as on the total block fees. Ordinal inscriptions increase the congestion on the Bitcoin blockchain, pushing overall fees upward and increasing miners' revenue. This is good news, because in the long run the Bitcoin blockchain will have to be supported by large transaction fees. Building additional application layers, like ordinals, have the potential to keep a high congestion.

One limit of our paper is that we only consider ordinal inscriptions and not further trades of the inscribed ordinals. After they are inscribed, ordinals can be transferred just like any other assets, but we do not consider these trades. These transactions would look like any other regular Bitcoin transactions. Although, it could be possible to track all inscribed ordinals, we do not think it will add much to the discussion about transaction fees. While most of the congestion comes from the inscriptions that can be large in size, we leave this for further studies.

A few months after Bitcoin ordinals started to be used, other standards were built on top of ordinals. Most notably, we note that the so called BRC-20 allows users to create and transfer fungible assets directly on the Bitcoin blockchain using ordinal inscriptions. The general idea is that each basic operation (creation, mint, transfer) of a token could be written in an ordinal inscription. BRC-20 started to be heavily used on April 20th 2023, after the end of our sample period. We leave this to further studies.

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Appendix: Tables

Table 7: Description of the variables of the database used in this paper

Name	S	Description
Panel A: Transaction level variables / Collected		
Transaction fee		Transaction fees in BTC of the transaction
Transaction size		Size of the transaction in <i>vbytes</i>
Transaction height		Block height of the transaction
Type		Type of the ordinal inscription (application, audio, image, model, text, video or other)
Panel B: Block level variables / Collected		
Block time		Timestamp of the block
Block height		Block height
Block size		Size of the block in <i>weight</i>
Block fees		Block total fees in BTC
Tx count		Number of transactions in the block
BTC price		Time weighted average of a per-minute volume-weighted average of bitcoin price at this block
Panel C: Block level variables / Computed		
Block fee per <i>vbyte</i>		Average transaction fee in BTC per <i>vbyte</i> for this block
Block USD fee per <i>vbyte</i>		Average transaction fee in USD per <i>vbyte</i> for this block
Ordinal count		Number of ordinal inscriptions in this block
Ordinal text count		Number of text ordinal inscriptions in this block
Ordinal non-text count		Number of non-text ordinal inscriptions in this block
Ordinal size		Size of ordinal inscriptions in this block in <i>vbytes</i>
Ordinal fees		Total fees of ordinal inscriptions in this block in BTC
Ordinal non-text size		Size of non-text ordinal inscriptions in this block in <i>vbytes</i>
Ordinal non-text fees		Total fees of non-text ordinal inscriptions in this block in BTC
Ordinal text size		Size of text ordinal inscriptions in this block in <i>vbytes</i>
Ordinal text fees		Total fees of text ordinal inscriptions in this block in BTC
Ordinal fees per <i>vbyte</i>		Average transaction fee in BTC per <i>vbyte</i> for all ordinal inscriptions in this block
Non-ordinal fees per <i>vbyte</i>		Average transaction fee in BTC per <i>vbyte</i> for all non-ordinal inscriptions in this block
Ordinal non-text fees per <i>vbyte</i>		Average transaction fee in BTC per <i>vbyte</i> for all non-text ordinal inscriptions in this block
Ordinal text fees per <i>vbyte</i>		Average transaction fee in BTC per <i>vbyte</i> for all text ordinal inscriptions in this block